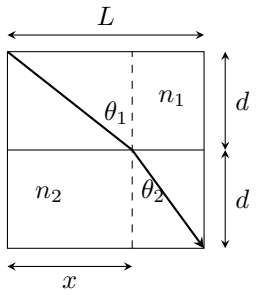
7.1.2 Fermat's Principle

[28] Ultimately, all fundamental laws of physics can be expressed using the calculus of variations. One of the first examples was a suggestion by Fermat, in 1662, that a light ray travels the path that takes the least time. This is known as *Fermat's principle*.

[29] For a light ray moving in a vacuum, Fermat's principle just reaffirms the obvious statement that light travels in a straight line between any two points. But things become more interesting when we consider light travelling in some material, where it no longer travels at speed c, but instead at the lower speed $v = c/n$ where n is called the *refractive index*.

[30] Suppose that a light ray moves between two different materials, the first with an index of refraction n_1, the second with n_2 as shown in the figure. The light ray travels a total distance L in the horizontal direction, and a distance $2d$ in the vertical direction, but we suppose that it has a choice about the point x where it transitions from one material to the other. We can think of the total time taken T to be a function of x: it is



$$T(x) = \frac{\sqrt{d^2 + x^2}}{v_1} + \frac{\sqrt{d^2 + (L-x)^2}}{v_2}.$$

[31] To minimize this, we don't need any fancy calculus of variations. We just need to differentiate

$$\frac{dT}{dx} = \frac{x}{v_1\sqrt{d^2 + x^2}} - \frac{L-x}{v_2\sqrt{d^2 + (L-x)^2}} = 0. \tag{7.20}$$

> **複習註解一：為什麼（7.20）要令 $dT/dx = 0$？**
>
> 起點與終點固定時，改變光線穿過介面的位置 x，就會改變兩段路程及總時間 $T(x)$。依費馬原理，本例要找出使總時間最小的交會位置。
> 若某位置滿足 $T'(x) \neq 0$，就能選擇向左或向右微調 x，使時間減少，因此該位置不會是極小值。故極小值必須滿足
>
> $$T'(x) = 0,$$
>
> 這就是（7.20）的由來：交會點微小移動時，第一段增加的時間與第二段減少的時間在一階近似下恰好抵銷。
> 單憑 $T'(x) = 0$ 只能判定駐點；本例另有
>
> $$T''(x) = \frac{d^2}{v_1(d^2 + x^2)^{3/2}} + \frac{d^2}{v_2[d^2 + (L-x)^2]^{3/2}} > 0,$$
>
> 所以此駐點確實給出最小時間。一般而言，費馬原理要求的是時間取駐值。

[32] This has a nice geometrical interpretation in terms of the angles shown in the figure: we have

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2} \quad \Longrightarrow \quad n_1 \sin\theta_1 = n_2 \sin\theta_2. \tag{7.21}$$

[33] This is *Snell's law* for refraction.

[34] In more complicated situations, Fermat's principle gives a problem that falls very much in the camp of calculus of variations. Suppose that light moves in the (x, y)-plane with the refractive index varying continuously as $n(y)$. An example of this happens for the air above a hot, shimmering road, where refraction causes a *mirage* to appear as light from the sky bends.

[35] We consider light travelling from $(x, y) = (a, A)$ to $(x, y) = (b, B)$. The time taken along a general path $y(x)$ is given by

$$T = \frac{1}{c}\int_a^b dx\, n(y)\sqrt{1 + y'(x)^2}. \tag{7.22}$$

複習註解二：（7.22）如何表示光沿曲線行進的時間？

$y(x)$ 描述從 (a, A) 到 (b, B) 的一條候選光路；折射率 $n(y)$ 隨高度改變，因此當地光速為

$$v(y) = \frac{c}{n(y)}.$$

書中的例子是熱路面上方的空氣，但公式也適用於其他具有折射率分布 $n(y)$ 的介質。
沿路徑由左往右取一小段，其弧長與所需時間分別為

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + y'(x)^2}\, dx, \qquad dt = \frac{ds}{v(y)} = \frac{n(y)}{c}\, ds.$$

把各小段的時間累加，即得

$$T[y] = \frac{1}{c}\int_a^b n(y(x))\sqrt{1 + y'(x)^2}\, dx.$$

（**7.22**）計算的是沿指定路徑行進的時間，尚未決定實際光路。接下來才對路徑作變分，利用費馬原理求出光所走的路徑。

[36] The Euler–Lagrange equation is

$$\frac{\partial L}{\partial y} - \frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right) = \sqrt{1+y'(x)^2}\,\frac{\partial n}{\partial y} - \frac{d}{dx}\left(\frac{n(y)y'(x)}{\sqrt{1+y'(x)^2}}\right) = 0. \tag{7.23}$$

[37] That's rather a mess. But there is, it turns out, a hidden simplification. You can show that the Euler–Lagrange equation (7.23) implies that

$$\frac{n(y)}{\sqrt{1+y'^2}} = \text{constant}. \tag{7.24}$$

複習註解三：如何由（7.23）得到（7.24）？

令 $n_y = dn/dy$。因為 $n = n(y(x))$，由鏈鎖律，

$$\frac{dn}{dx} = n_y y'.$$

因此

$$\frac{d}{dx}\left(\frac{ny'}{\sqrt{1+y'^2}}\right) = \frac{n_y y'^2}{\sqrt{1+y'^2}} + \frac{ny''}{(1+y'^2)^{3/2}}.$$

代回（7.23）並合併前兩項，可得

$$\frac{n_y}{\sqrt{1+y'^2}} - \frac{ny''}{(1+y'^2)^{3/2}} = 0,$$

亦即

$$\boxed{n_y(1+y'^2) - ny'' = 0.}$$

接著，對（7.24）左邊微分：

$$\begin{aligned}\frac{d}{dx}\left(\frac{n}{\sqrt{1+y'^2}}\right) &= \frac{n_y y'}{\sqrt{1+y'^2}} - \frac{ny'y''}{(1+y'^2)^{3/2}} \\ &= \frac{y'}{(1+y'^2)^{3/2}}\underbrace{\left[n_y(1+y'^2) - ny''\right]}_{=0}.\end{aligned}$$

故其導數為零，得到

$$\boxed{\frac{n(y)}{\sqrt{1+y'^2}} = \text{constant}.}$$

這表示沿著滿足 **Euler–Lagrange** 方程的光路，折射率 $n(y)$ 與斜率 y' 雖可改變，但上述比值保持不變。推導過程沒有除以 y'，所以在 $y' = 0$ 的位置也適用。

這個組合也有系統性的來源：當 L 沒有顯含 x 時，沿著 Euler–Lagrange 方程的解，$L - y'\partial L/\partial y'$ 為常數。本例

$$\begin{aligned}L - y'\frac{\partial L}{\partial y'} &= n\sqrt{1+y'^2} - \frac{ny'^2}{\sqrt{1+y'^2}} \\ &= \frac{n}{\sqrt{1+y'^2}},\end{aligned}$$

正好就是（7.24）的左邊。

[38] Showing this directly is straightforward, but laborious: just use the chain rule to get an explicit expression for that d/dx term in the Euler–Lagrange equation (7.23) and then show that the same Euler–Lagrange equation appears when you differentiate the combination $n(y)/\sqrt{1+y'^2}$. Later in this chapter, we'll give a more direct way to see why (7.24) is constant. (It is related to the conservation of energy in classical mechanics.)

[39] For a given profile $n(y)$, we can go ahead and solve (7.24). For example, we take the refractive index to be

$$n(y) = \sqrt{a - by}. \tag{7.25}$$

[40] Then (7.24) becomes

$$\frac{a-by}{1+y'^2} = k^2 \quad \Longrightarrow \quad y'(x)^2 = \frac{b}{k^2}\big(y_0 - y(x)\big). \tag{7.26}$$

[41] Here k is the constant on the right-hand side of (7.24) and $y_0 = (a - k^2)/b$. The solution to this equation is

$$y(x) = y_0 - \frac{b}{4k^2}(x - x_0)^2. \tag{7.27}$$

[42] We see that, with the refractive index (7.25), light doesn't travel in a straight line at all. Instead, it travels in a parabola! That's rather striking, not least because a ball thrown under the influence of gravity also travels in a parabola. It might make you wonder if perhaps we can phrase more general mechanics problems in a similar way. And, indeed, we can. Although, as we'll see shortly, it's typically not the time that we want to minimise.